The Memory That Crossed Worlds

Omega-01 | 1,800 held-out episodes | 20 paired seeds | Five treatments

Preserved memory showed no clear success advantage.
Preserved success was 75.0% and standardized success was 71.4%. The paired difference was +3.6 percentage points, with a 95% seed interval of -3.1 to +10.3.

Why the 1.4 actions matched

Across 432 preserved decisions, the original adapter made 14 semantic lookups and zero queries to old Earth keys. The old locations were stored outside the neural input path. Removing or shuffling archived entries changed zero actions and zero input vectors. Books were reachable on 21 ticks and were never selected.

Held-out results

Treatment	Success	Cold Brier	Final Brier	Reads	Energy / trial
Preserved	75.0%	0.218	0.204	84	1.50
Standardized	71.4%	0.250	0.231	95	1.94
Relevant removed	71.4%	0.250	0.231	95	1.94
Context shuffled	73.9%	0.240	0.215	89	1.75
Rule-based reference	74.2%	0.250	0.224	95	1.92

Brier error measures the probability forecasts for both choices. Lower is better. Cold estimates precede practice. Final estimates include any evidence the agent chose to acquire.

What the bridge did

All neural arms used the same verified age-944 checkpoint state. The bridge scored 37 real archived observations using generic properties such as availability, reported availability and energy cost. It kept old names and locations out of the new observation path. The archive ledger was reconstructed from recordings and the similarity rule was supplied by the experiment.

Decisions, books and causal checks

Observed effects of evidence and single-record removal

Success by scenario

Task	Preserved	Standardized	Relevant removed	Shuffled	Reference
Resource availability	77.5%	78.3%	78.3%	76.7%	79.2%
Delayed effort cost	77.5%	70.0%	70.0%	79.2%	75.8%
Reported availability	70.0%	65.8%	65.8%	65.8%	67.5%

Removing one retrieved record changed 1 of 60 replayed object choices. Each replay kept the same scene, body and learned local state. In scene 14201420, removing a verified Earth report from tick 746 switched Omega from the successful object to the failing one.

Independent use of documents

Preserved Omega chose to read 84 times. The book was a reachable object, cost energy and time, and sometimes contained incomplete or wrong entries. The preserved arm checked 79 evidence claims against its own chosen outcomes across reading and physical inspections. Choosing a supplied Read action establishes evidence seeking under the menu.

In effort scene 14200414, both mechanisms initially scored 0.763. The folio raised the pulley lever to 0.917. Omega switched from the gear press and stayed within the delayed energy budget.

Scope and compute

Mean preserved CPU time was 18.0 ms per held-out episode. All arms used the same grid body, task cues and modeled comparison budget. The control keeps physics fixed and the experiment runs offline. The result is specific to the supplied bridge and three task families. The workshop's delayed energy charge was introduced in this study. The old energy records concern observed effort costs.

Sources Baseline memory_access.jsonl and audit.json, locked results.csv, paired.csv, archive_records.json, counterfactuals.json and decision traces.